POWER AND FILTERING

5V
C16 10uF
GND

+3V3
C17 10uF
GND

LEDS

+3V3
D2 LED
R16
C2913417
GND

+3V3
C3 10uF
GND

IO1
D3 LED
R25
R24
C2913417
GND

MCU

+3V3
C39 1uF
C40 10nF
R31 10K
GND GND
VDD_CPU
IO45
C10 0.1uF
GND
+3V3
+3V3
500mA
L3 22uH
C25 10uF
C26 1uF
C18 0.1uF
C19 0.1uF
GND GND GND GND
VDD_CPU
VDD_SPI
VDD_CPU
+3V3
C11 0.1uF
GND
VDD3
VDD3P3_RTC
VDD_SPI
VDD3P3_CPU
55 2 20 29 46

EN
C2888884 Reset1 Reset C318884
GND
+3V3
R32 10K
IO0
C2888884 Flash1 Flash C318884
GND

C2913196

IO0 IO1 IO2 IO3 IO4/BAT IO5 IO6 IO7 IO8 IO9 IO10 IO11 IO12 IO13 IO14 IO17/SCL IO18/SDA USB_D- USB_D+ IO21
GPIO0 GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6 GPIO7 GPIO8 GPIO9 GPIO10 GPIO11 GPIO12 GPIO13 GPIO14 GPIO17 GPIO18 GPIO19/USB_D- GPIO20/USB_D+ GPIO21

XTAL_32K_P XTAL_32K_N XTAL_P XTAL_N

SPICS1 SPIHD SPIWP SPICS0 SPICLK SPIQ SPID SPICLK_N SPICLK_P
R30 SPIHD
R29 SPIWP
R28 SPICS0
R27 SPICSK
R26 SPIQ
SPID
IO48
IO47

GPIO33 GPIO34 GPIO35 GPIO36 GPIO37 GPIO38 MTCK MTDO MTDI MTMS U0TXD U0RXD GPIO45 GPIO46 CHIP_PU LNA_IN
IO33 IO34 IO35 IO36 IO37 IO38 IO39 IO40 IO41 IO42 TX RX IO45 IO46

GND
U1 ESP32-S3
GND

GND GND
12pF C4
40MHz(10ppm) Y2
12pF C33
GND
R23 2K

C32 0.1uF
C13 1uF
VDD_SPI
GND GND

Y1 32.768 kHz
XTAL32N XTAL32P
C37 12pF
C38 12pF
GND GND

IO45/IO46: pulldown-

+3V3
C2888884 R22
EN
AE1 Antenna
C15 0.1uF
GND
L2 4.3nH
DNP1 15pF
C31 15pF
L5 DNP
C14 DNP
GND GND GND

jst-xh

VOUT_SW IO38
J2 LED1
GND
VOUT_SW IO38
J3 LED2
GND
VBAT
J4 BAT
GND
VOUT_SW VOUT VIN
J5 B4B-XH-A_LF__SN_
GND

MEMORY

VDD_SPI
SPICS0 SPISCK SPID SPIQ SPIWP SPIHD
CS CLK DI/IO0 DO/IO1 WP/IO2 HOLD/RESET/IO3 GND VCC
C97521
U2 W25Q128JVS
GND

USB-C

J1
USB_C_Receptacle_USB2.0_14P
C2765186
VBUS CC1 CC2 D- D- D+ D+ SHELL GND S1 A1
A4 B5 A7 B7 A6 B6
5V
R33 5.1K
R34 5.1K
C24 5pF
GND
USB_D-
USB_D+
R35 22
R36 22
USB_D-
USB_D+
5V 5V
C6 10uF
C22 0.1uF
GND GND

VOLTAGE REGULATOR

VSYS
U4
AMS1117-3.3
VIN VOUT GND
C7 10uF
GND
+3V3
C20 10uF
C35 0.1uF
GND GND
+3V3 +3V3
GND GND1

BATTERY CHARGER

U5
IP5306
VIN KEY VOUT LED1 LED2 LED3 SW BAT POWERPAD
VIN
R3 2R
C1 10uF
C5 10uF
GND
GND
C9 22uF
C12 22uF
C21 22uF
VOUT
LED D5 LED D6 LED D7 LED D8
L1 1uH
GND
10uF C2
10uF C23
R1 2R
VBAT

5V
R20 5.1K
R21 5.1K
GND
D4 SS14
VSYS
Q1 AO3401A
VBAT
C8 10uF
C25 22uF
C26 22uF
IO4/BAT
GND

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